

A Rule-Constrained Large Language Model Pipeline for Patient-Friendly Interpretation of Vietnamese Blood Test Results

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1. The problem: having a result without understanding it

A patient leaves the laboratory with a panel of numbers and terms:

PDW

MCHC

Monocyte

G/L

fL

pg

The reference interval is printed beside each value, so a patient can see a number falls outside it, but not what it measures or what to do next.

60.3%

of older adults in Da Nang have inadequate
general health literacy

46%

of patients search online to understand
their own test results

Sources: cross-sectional study in Da Nang (n=204); Giardina et al., JAMIA 2018 (n=93)

2. What the patient actually receives

BLOOD TEST REPORT · COMPLETE BLOOD COUNT			
Report no. 2026-000123 · Illustrative data, not a real patient record			
Test	Result	Unit	Reference range
WBC	11.8 ↑	G/L	3.4 – 9.4
NEUT%	78.5 ↑	%	40.9 – 73.7
LYMPH%	14.2 ↓	%	17.2 – 46.1
MONO%	6.1	%	3.7 – 10.9
RBC	4.92	T/L	4.37 – 5.69
HGB	145.6	g/L	130 – 175
HCT	44.1	%	38.8 – 50.0
MCV	89.6	fL	80 – 100
MCHC	330	g/L	320 – 360
RDW	15.8 ↑	%	11.5 – 14.5
PLT	118 ↓	G/L	150 – 450
PDW	17.2	fL	10 – 18
MPV	9.4	fL	6.5 – 12.0
The reference range is printed, so the patient sees which value is outside it, but not what the analyte measures, or what to do next.			

Illustrative report. Red rows are outside the reference interval printed beside each value.

3. How patients currently close the gap

They photograph the report and ask a general-purpose language model.

Prompt *"Briefly explain what white blood cells are" (in Vietnamese)*

3B model response

*"White blood cells ... are lymphocytes capable of moving through the blood ...
They help the body fight viruses, bacteria and cancer-causing cells.
When the immune system is weakened, white blood cells also decrease ..."*



Definitional error

Lymphocytes are ONE type of white blood cell, not a synonym



Unprompted disease

Raises cancer and immune deficiency without being asked

A patient with an entirely normal result meets one definitional error and two alarming diseases in a single short answer.

4. But the most dangerous failure runs the other way

Terminological error

Visible to the reader
and invites verification

False reassurance

Fluent and comfortable,
and removes the reason to seek care

False reassurance: asserting that all results are normal on a panel that is NOT.

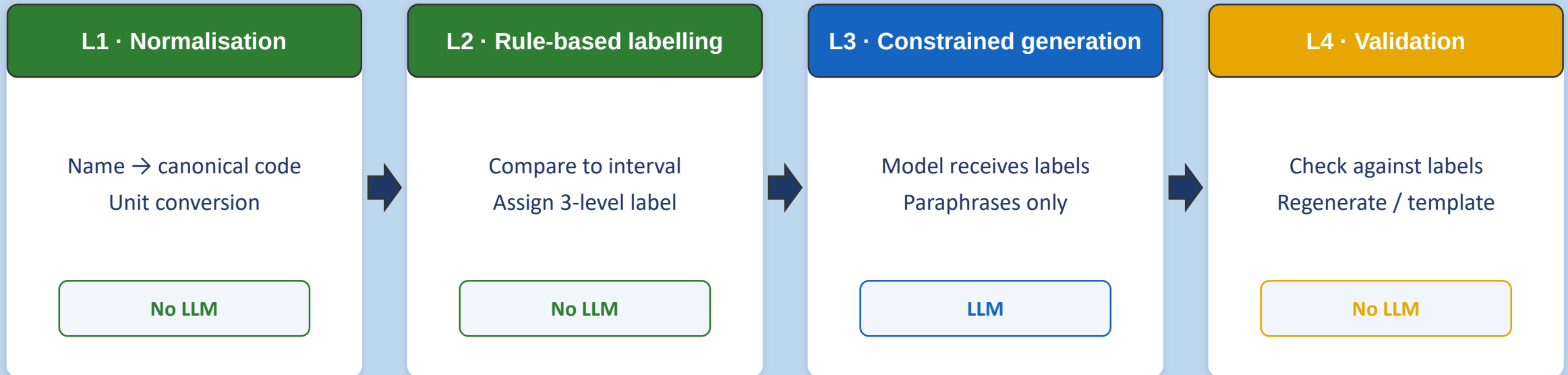
An actual output from our experiment, on a panel labelled "seek care":

"All test results are within the normal range."

We treat this as a distinct error class and measure it separately.

5. Four-layer architecture

Principle: NO clinical judgement is delegated to the language model



Failing output → regenerate → fixed template after the retry limit. Unverified text is never displayed.

6. Layer 1 — Name and unit normalisation

One analyte, many names

WBC

SLBC

Số lượng bạch cầu

* WBC

The % and # markers distinguish percentage from absolute count, and may precede (% Neu) or follow (NEUT%) the name.

Units are the serious problem

Hb = 14.56 g/dL

Reference interval: 130-175 g/L

✗ Unconverted → classified as "SEEK CARE"

✓ Converted → 145.6 g/L → Normal

Rule: unknown or missing unit → REFUSE to process. Never guess.

External evidence: 6–19% of laboratory tests cannot be accurately mapped to LOINC; an analysis of 163 million results identified 2,019 distinct unit expressions.

7. Layers 3 & 4 — Constrained generation and validation

Layer 3 · Three constraints

Input

The model is GIVEN labels, never asked to derive them

Content

No disease names, causes, medications or dosages

Output

Fixed JSON schema required

Layer 4 · Six error classes checked

- Does the stated direction match the Layer 2 label?
- Does the prose contradict the assigned label?
- Are forbidden terms present?
- Is any analyte fabricated?
- Is any analyte omitted?
- Does the closing statement falsely reassure?

Model: Qwen2.5-3B served locally via Ollama. Data never leaves the institution, and no GPU is required.

8. Evaluation design

Dataset

- 150 synthetic complete blood count panels
- 10–18 analytes per panel
- Equally split across 3 severity classes
- Random aliases and units per analyte

Why synthetic data

- Ground truth labels required; real reports carry none
- Balanced design required; real panels are mostly normal
- No patient data handled

Three configurations

baseline

Raw values given to the model. No labels, no constraints, no validation.

constrained

Layers 1–3. Labels and constraints, but output not verified.

full

Layers 1–4. Complete, with regeneration and template fallback.

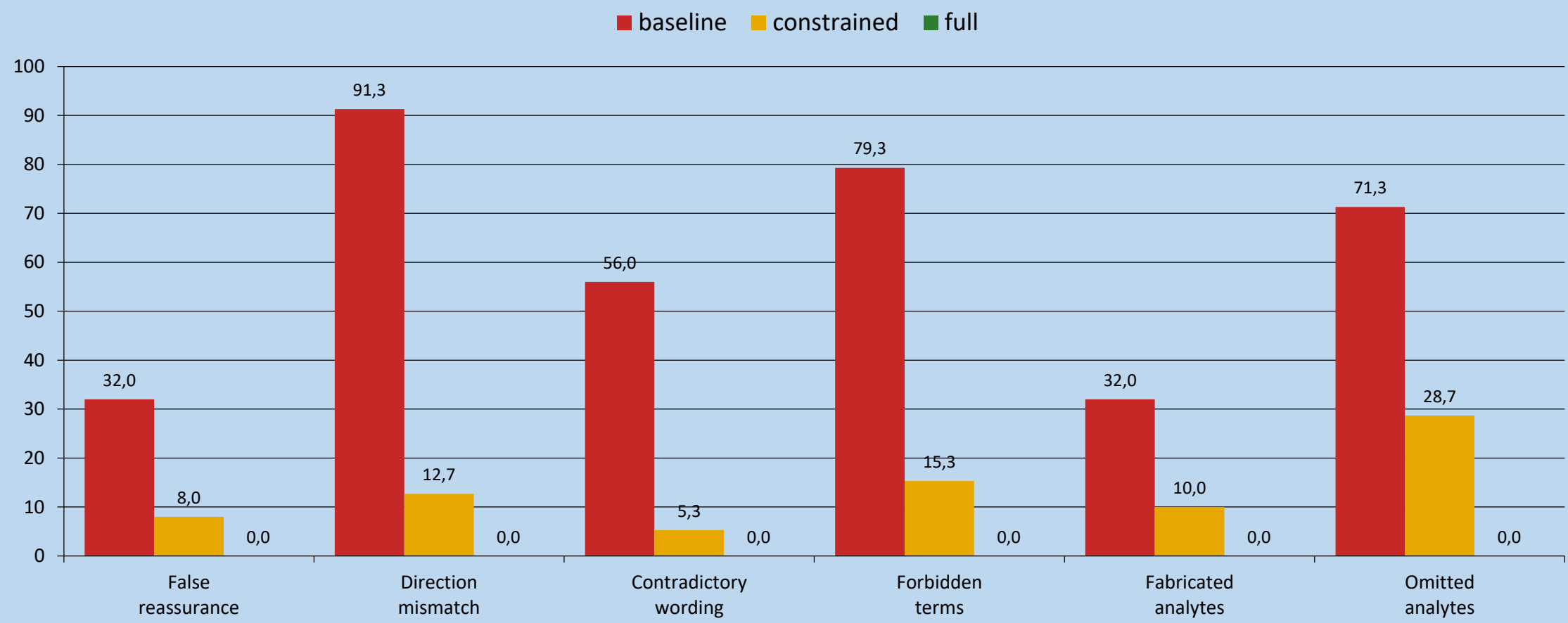
9. Results

Configuration	Safe output	Model pass	False reassur.	Direction mismatch	Contradict. wording	Forbidden terms	Fabricated analytes	Omitted analytes
baseline	0.0	0.0	32.0	91.3	56.0	79.3	32.0	71.3
constrained	48.0	48.0	8.0	12.7	5.3	15.3	10.0	28.7
full	100.0	64.0	0.0	0.0	0.0	0.0	0.0	0.0

Values: % of panels · n = 150 · Lower is better for all error columns

Every error metric decreases monotonically across the three configurations.

10. The same results, as a chart



Vertical axis: % of panels with that error. Lower is better. Full is zero in all six classes.

11. Three key findings

32% → 0%

False reassurance

Unconstrained: roughly one patient in three with abnormal results would be told nothing needed attention.

91.3% → 0%

Direction mismatch

The model judging for itself fails almost always. Moved to a rule layer, classification becomes exact.

48%

Constraints insufficient

Over half of prompt-constrained outputs still failed validation. Instruction does not guarantee compliance.

The 100% comprises 64% model-generated text and 36% fixed templates. That guarantee comes from the pipeline, and is not evidence that a 3B model performs this task reliably.

12. Discussion — two observations

1

False reassurance deserves its own metric

Current evaluations report readability and factual accuracy. A falsely reassuring sentence may still score HIGH on readability, because it is simple and calming. The harm is that it stops the reader from seeking care.

2

Unit standardisation is a safety precondition

The unit error occurred inside the rule layer, the layer treated as exact. No downstream validation detects it, because every later layer faithfully reproduces a confidently wrong label.

Data standardisation (LOINC, HL7 FHIR) is therefore a precondition for rule-based clinical logic, not only a data-reuse concern.

13. Limitations

No clinical validation

All evaluation is automated. No medical professional reviewed the generated explanations.

Thresholds not grounded

Severity factors separating "monitor" from "seek care" were set heuristically, not against a critical value list.

Synthetic data

Real reports carry noise, missing fields and formatting variation this study does not capture.

Narrow scope

Complete blood count only. No biochemistry, one-sided intervals, qualitative results, or OCR.

14. Conclusion and future work

The language model is not permitted to judge whether a value is abnormal. Rules decide first, the model paraphrases, a validator checks one against the other.

32% → 0%

false reassurance

100%

safe output

0

GPUs required

Future work, in order of importance

- **Clinical validation by medical professionals (most pressing)**
- Replace heuristic thresholds with a critical value list
- Extend to biochemistry and qualitative results
- Speech output layer for elderly and low-literacy patients
- Evaluation on real reports from multiple laboratories

Thank you

Questions and discussion

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